CS 1143-TW01S: Beginning Programming Syllabus

Business & Information Technology (BIT), Spring 2023

Instructor Information

Name: Tashfeen, Ahmad
Class Time & Location: Mondays 5:30-8:00PM in Library 122.
Office Hours: Wednesdays 2:30-5:30PM and 8:00-9:00PM (or by appointment) in LIB 314 or Zoom. Fridays 6:00AM-11:00AM online over Zoom. If I am not answering Zoom, I am seeing an in-person student.
Email: ahmad.tashfeen@occc.edu, please prefix "[CSXXXX-XXXXX]" to all email subjects so they don't get lost.
Phone (BIT Office): (405) 682-7888
Text: Gaddis, Tony. *Starting Out with C++: From Control Structures Through Objects*. Pearson, 2018. 9th Edition, Brief Version. ISBN-13: 9780134996042.
Equipment, software etc.: A working computer, internet connection, a text editor, e. g., Sublime Text or Emacs, GNU Compiler Collection (GCC) and Windows Subsystem for Linux (WSL) for Windows users. Please contact the instructor (email preferred) if you are having difficulty acquiring any of the course materials. Further instructions will be provided for GCC and WSL.

College Resources

College Website: http://www.occc.edu
Student Handbook: https://www.occc.edu/catalog
Student Computer Center: https://www.occc.edu/bit/scc
Online Student Resources: https://www.occc.edu/online/resources
Student Support Services: 405-682-7520 and https://www.occc.edu/support
Email Address Request: For more information https://www.occc.edu/c4lt/email. It is each student's responsibility to check this often and it is important to do so. I will distribute important information and updates throughout the semester via email.

Policies and Procedures

Disclaimer: Information hereinafter, e. g., the assignments, point values, schedule, and testing dates are subject to change in the event of extenuating circumstances or if class progress warrants. Adjustments may be made as the semester progresses.
Academic Integrity Policy: The absence of academic integrity is described as cheating, generally defined as the deception about one's work to others. Any violations to the academic integrity policy will be addressed according to Academic Integrity college policy (Policy 4016). For examples and to learn more visit the link below:
https://www.occc.edu/catalog/regulatory/academic-information/integrity
Late Work and Due Dates: All assignments are due by the posted due date. Due dates will be posted as and when the assignments are assigned. When accepted, work that is $n \leq 7$ days late will receive no more than $100(1 - 0.1n)\%$ of credit. That is 10% deduction for each day late. For certain assignments and tasks, late work will not be accepted.
Grading Criteria and Points: There are a total of 1000 possible points. 300 points for unit exams, 230 for programming assignments, 180 for *MyProgrammingLab*, 100 for chapter quizzes, 90 for practice quizzes, 100 for programming quizzes. At least 900 points are required for an A, 800-899 points for B, 700-799 for C, 600-699 for D (does *not* satisfy CS pre-requisite grade requirements) and an F otherwise.
Communication and Grades: I will usually respond to emails within a business day. Most work should be graded within a week after the posted due date on average.
Unit Exams: There are four Unit Exams. These are handled in the same way as the Chapter Quizzes; you take them from home. They are not timed. The instructor does not require the use of any "proctoring" software or a "lockdown" web browser.

Programming Assignments: You will do a lot of programming assignments, as that is how you learn to program. Assignments and due dates are given on Moodle. Most come from the textbook end-of-chapter pages (blue pages) with additional specifications given on Moodle. Do *not* use techniques from chapters not yet covered to complete the homework. Do not use language features that are not covered in this class on homework assignments.

MyProgrammingLab: *MyProgrammingLab* assignments are completed at http://myprogramminglab.com. MyProgrammingLab assignments are due with the Unit Exams and will not be accepted late. Note that you can earn some extra credit here; do problems in excess of the assigned number and pick up some extra points!

Chapter Quizzes: Chapter Quizzes are due at the end of each chapter. These are timed; You get one try and ten minutes so run through the Practice Quizzes a few times first to prepare for these.

Practice Quizzes: Practice Quizzes draw from the same question bank as the Unit Exams and the Chapter Quizzes. You can take the Practice Quizzes as many times as you like; each time you will (probably) see different questions, and the high grade stands. So that you can continue to practice for the exams, Practice Quizzes stay open until the Unit Exams are due.

Programming Quizzes: Starting with Chapter 5 you will start to see Programming Quizzes, in which you write a program (usually just a modification of a homework problem) to read and process a file, and then answer questions about the results. You generally get three tries on these.

Course Description: Designed for Computer Science majors, this course affords students a basic understanding of computer programming. Students will utilize accepted programming concepts and perform number system conversions and arithmetic. In addition, they will design and code structured modular programs using design tools such as hierarchy charts, flowcharts, and pseudocode.

Course Competencies/ Objectives: The core competencies of this course are algorithm design and structured modular programming. The theoretical concepts introduced in this class are universal. In addition, many of the programming languages you will encounter are part of the "C Family" of languages; the data types and control structures you will master in this class using C++ are common to many languages. Students who complete this course will be able to:

- Name and describe the purpose of the fundamental components of computing hardware.
- Understand the relationships between the number systems used in computing and perform conversions between those number systems.
- Analyze problems and create solutions to those problems by developing flowcharts or pseudocode that utilize the fundamental control structures (sequence, iteration, and selection) and then implement those solutions in C++.
- Design, input, compile, debug, and test modular C++ programs that make use of the primitive data types as both variables and constants, arithmetic and relational operators, input and output operations using both the console and files, control structures, arrays, functions, and text files.

Assessment of Student Learning: Oklahoma City Community College is committed to providing quality educational experiences to all students and to strive for continuous improvement in its programs and services. Student learning assessment is vital to the educational process and can be of significant value to you and to the students who follow you as well as contribute to their education success. To ensure that adequate assessment information is available to allow OCCC to continuously improve programs and services, students may be asked to participate in personal interviews; take program and/or general education assessments, which could be to take tests; give oral presentations, write assignments, take surveys, or engage in other activities. You may be asked to complete the assessments, tests, and other activities during a designated time, which may include class periods in a semester. These opportunities are your chance to help OCCC improve the courses, programs, and services which could affect you and will certainly impact students in the future.

Class Requirements: The on-campus class will be taught in a computer-equipped classroom. Online students are recommended to complete the work for each week as laid out in Moodle. Class material will be presented on-line. It is the student's responsibility to check for new assignments, class announcements, due dates, etc. Completed work will be submitted on-line through Moodle. Both campus and online students can have access to the Student Computer Center. It is your responsibility to arrange your

schedule for completion of assignments. It is of utmost importance that you stay current and turn all work in on time.

Learning Support: Student success is a high priority in the BIT Division. In case the instructor feels that a student might benefit more with personalized learning, student names may be referred to Student support Services. More details can be found here**.**

On Campus Classes: In respect for the instructor and fellow students, cell phones, pagers, etc. are to be turned off or placed on vibrate during the class. If there is an emergency requiring availability via a cell phone/pager, please let the instructor know at the beginning of that class period. Students may not surf the net, check email, text or engage in similar activities during class time. Not only does it affect the person doing it, but it is a potential distraction to others in the class.

Issuance of Grades: Homework and exams are typically graded within one week of the due date. Grades will be posted in Moodle at http://online.occc.edu. Check the grades area after each assignment has been graded. Students have one week after the grades have been posted to challenge a grade. Failure to request the review in a timely manner or to provide requested information will result in the grade remaining as originally posted. It is highly recommended that all graded assignments (for all classes) be saved until after the final grade has been posted.

Assistance with Assignments: In addition to my being available during office hours and via email, the Student Computer Center staff and tutors are also available for assistance. While some of the best learning takes place when we solve problems ourselves, please do not wait until it is too late. Ask for help when you need it. They will be happy to explain concepts and procedures. A list of the full-time staff can be found at the following URL. There are also other tutors and supplemental instructors available depending on the semester. Check with the Student Computer Center for details. Information can be accessed at http://occc.edu/bit/scc.html.

Institutional Policies

Network Acceptable Use Policy: Every user of the college network, equipment, communications systems, and/or college-owned software will comply with the Information Technology Resources Acceptable Use Policy. Information regarding the policy can be found at https://www.occc.edu/wp-content/legacy/policy/pdf/3058.pdf.

Accommodation Statement: Oklahoma City Community College complies with Section 504 of the Rehabilitation Act & the Americans with Disabilities Act. Students with disabilities who seek academic adjustments/accommodations must make their request by contacting the office of Student Accessibility and Support located on the first floor of the main building near SEM entry 3 or by calling 405-682-7520. All academic adjustments/ accommodations must be approved by Student Accessibility and Support. If you have been approved by Student Accessibility and Support to receive academic adjustments/accommodations you will receive an email from them with your approved accommodations attached. You should forward that email to me so I will know what is authorized and then you must talk with me either after class or during my office hours. This will allow me to be better informed on how to assist you with access during the semester. To respect your privacy I will not approach you, but the academic adjustments/accommodations must be discussed to ensure ideal implementation for you. All information will remain confidential.

Pregnancy and Parenting Statement: Oklahoma City Community College does not discriminate against any student, or exclude any student from its education program or activity, including any class or extracurricular activity, on the basis of parenting, such student's pregnancy, childbirth, false pregnancy, termination of pregnancy, or recovery therefrom. For the purpose of this policy Caretaking is defined as a person providing for the needs of a child and Parenting is defined as the raising of a child by the child's parents in the reasonably immediate post- partum period. Oklahoma City Community College has a new policy (Policy 1013) with procedures for ensuring the protection and equal treatment of pregnant individuals, persons with pregnancy-related conditions, and new parents. Students are encouraged to work with their faculty members and OCCC's support systems to devise a plan for how to best address the conditions as pregnancy progresses, anticipate the need for leaves, minimize the academic impact of their absence, and get back on track as efficiently and comfortably as possible. If

you should need assistance you can contact the Equity and Compliance Coordinator, Kimberly Rundell, at Kimberly.m.rundell@occc.edu or 405-682-7850. You can learn more about Reasonable Accommodation of Students Affected by Pregnancy, Childbirth, or Related Conditions and Authorization for Modified Academic Responsibilities Policy for Parenting Students at https://www.occc.edu/wp-content/legacy/policy/pdf/1013.pdf

Title IX: The Equity and Compliance Coordinator will assist with Title IX policy and procedures as needed. If you should need assistance you can contact the Equity and Compliance Coordinator, Kimberly Rundell, at Kimberly.m.rundell@occc.edu or 405-682-7850. You can also contact the Title IX Coordinator, Regina Switzer, at rswitzer@occc.edu or 405-682-7890.

Non-discrimination Statement: As a recipient of federal financial assistance for education activities, the College is required by Title IX of the Education Amendments of 1972 to ensure that all of its education programs and activities do not discriminate on the basis of sex/gender. Sex includes: sex, sex stereotypes, gender identity, gender expression, sexual orientation, and pregnancy or parenting status. The College also prohibits retaliation against any person opposing discrimination or participating in any discrimination investigation or complaint process internal or external to the institution. Sexual harassment, sexual assault, dating and domestic violence, and stalking are forms of sex discrimination, which are prohibited under Title IX and by College policy. Any member of the campus community, guest, or visitor who acts to deny, deprive, or limit the educational, employment, or social access, opportunities and/or benefits of any member of the College on the basis of sex is in violation of OCCC College Policy No. 1012.

Assessment of Student Learning: Oklahoma City Community College is committed to providing quality educational experiences to all students and to allow striving for continuous improvement in its programs and services. Student learning assessment is vital to the educational process and can be of significant value to you and to the students who follow you the student as well as contribute to their education success. To ensure that adequate assessment information is available to allow OCCC to continuously improve programs and services, students you may be asked to participate in personal interviews; take program and/or general education assessments, which could be tests; give oral presentations, write assignments, take surveys, or engage in other activities. You may be asked to complete the assessments, tests, and other activities during a designated times, which may include class periods in a semester. These opportunities are your chance to help OCCC improve the courses, programs, and services which could affect you and will certainly impact students in the future.

Attendance Requirements: OCCC expects students to attend class on a regular and punctual basis. If a student is absent from class, regardless of the cause, it is the responsibility of the student to communicate with the instructor to discuss work missed. The instructor will determine whether or not the student will be permitted to make up work and will decide on the time and nature of the make-up. For face-to-face courses, students are expected to attend each session. If you cannot attend class, please notify the instructor as soon as possible. For in-person classes, *any more than one* not excused absence will result in an automatic final grade of F (zero) in the class. For online and remote courses, students are expected to log in to the Moodle course section at least once per week. If for some reason you are unable to connect to Moodle, please contact the instructor and Campus Technical Support at 682-1611 extension 7777.

Never Attended-Non Attendance: Students who do not attend class(es) for which they have enrolled, will be reported as Never Attend and will receive a status of NA. The deadline for students to attend class or engage in an online class is the drop with refund date for the respective term.

- The NA will not be reported on the student's transcript.
- The status of NA is given regardless of the student's financial aid status and/or if the student has made any payments.
- A first semester student who is NA'd from all of their respective classes, will not be charged the $30 enrollment fee.
- Students will not be charged for the class(es) for which they receive an NA.
- Students' financial aid awards will be recalculated to remove NA courses, which may result in a reduced award amount.

Withdrawal Policy: A student choosing to withdraw from a class must complete and submit an add/drop form which may be obtained from the advising office or withdraw online through his/her student account before the beginning of the fourth quarter of a semester (i.e., through the 12th week of a 16-week semester or the sixth week of an eight-week semester). Students who do not take the initiative to officially withdraw from a class may be give the grade of "F" and will be responsible for full payment of tuition and fees for the class. If you have questions regarding this notice, please consult your instructor.

FERPA statement: The educational rights and privacy of students of Oklahoma City Community College are governed by the Family Educational Rights and Privacy Act (FERPA) of 1974 as amended. The Act ensures students certain rights with regard to their education records. Students are ensured: the right to inspect their education records; the right to request the amendment of those records if they believe they are inaccurate, misleading, or otherwise in violation of the student's privacy or other rights; the right to consent to the release of personally identifiable information, except to the extent that FERPA authorizes disclosure without consent, i.e., directory Information; the right to file a complaint with the U.S. Department of Education concerning alleged failures by Oklahoma City Community College to comply with the requirements of FERPA; if you wish to review any academic record, please approach your faculty member to arrange a mutually agreeable meeting day, time and location; when distance is a barrier for a face to face meeting, an electronic form of communication for a meeting will be arranged; students are encouraged to seek a review of their academic records while the course is in progress or shortly thereafter to ensure the record is maintained and available for review. The educational rights and privacy procedure of Oklahoma City Community College is published in the OCCC College Catalog and is available upon request from the OCCC Registrar's Office. For more information regarding your student rights under FERPA, visit http://www.occc.edu/records/FERPA.html.

Moodle statement: Moodle grades are provided for informational purposes only. In the event of a discrepancy between grades reported in Moodle and grades calculated according to the course syllabus, the latter of the two is the official grade for the class.

Copyright: Lectures and course materials in this course, including presentations, tests, exams, outlines, and similar materials, are protected by United States copyright law (Title 17, US Code). During this course, students are encouraged to take notes and download course materials for their own educational use. Students are prohibited from disseminating copyrighted course materials to any other person or entity, other than sharing them directly with another student enrolled in the course for purposes of studying, without prior written permission from the course instructor.

OCCC Emergency Procedures: The health and safety of all our students, faculty, and staff are OCCC's prime concern. The procedures outlined below are designed to deal with emergencies of various types. Students should always follow the lead of their instructors.

Fire: First notification will come from the fire alarm horns, sirens, and strobes. The class should gather their belongings, exit the building using the nearest exit, and move to a parking lot. *Do not use the elevators.* No alarm should be treated as a false alarm. Horns, sirens, and strobes are only used for fire alarms.

Fire (Special Considerations): If someone in your area is not physically capable of descending the stairwell, please ensure that they remain in the *"area of safe refuge"* located just inside each upper-level enclosed first stairwell. There are emergency phones located near each of these areas.

Medical Emergencies: For all medical emergencies use the "*Emergency"* button located on each classroom or office phone. The classroom or office phone will display your current location, allowing for fast response to your location. You can also report a medical emergency to the Campus Police Department using any phone by dialing 682-7872. Dialing 911 directly may delay emergency responders. Campus Police and Security Officers are Basic Life Support capable and will also direct necessary Emergency Medical Services to your location. Remember to treat all bodily fluids as if they were contaminated and a potential biohazard.

Bomb Threat/Evacuation: If you receive a bomb threat, document as much information as possible and push the *"emergency"* button on the phone. If the decision to evacuate is given, the phone will *sound* an alarm and *display* a text message. The class should gather their belongings, exit the building using the nearest exit, and move away from buildings to Faculty Circle by the most direct and safe route. Please

turn off *all* wireless devices. (Cell phones, radios, laptops, and other portable devices.) Updated information will be communicated as soon as it is available. Do not return to any building until there is an “All Clear” message.

Severe Weather: Tornado warnings with a potential impact to OCCC properties will be sent directly to the classroom phone. The phone will *sound* an alarm and *display* a text message. The class should gather their belongings, move away from exterior glass and exits, and move to safer areas. Designated for the building you are occupying. These “Safer Areas” are lower-level interior classrooms and hallways. Some buildings include stairwells as safer areas. You should familiarize yourself with the safer areas near your classroom(s). If the city/county sirens are sounding and OCCC is *not* in the warning area a message will be sent to the classroom phone advising this information.

Disturbance/Threats: If someone is causing a disturbance in a classroom call the Campus Police Department immediately. Use the “emergency” button located on each classroom phone. Distance yourself from that person, do not place yourself in the person’s exit path and remove all potential weapons from the area.

Shelter in place: If there is an armed person or active shooter on campus, the Campus Police will initiate a Shelter in Place alert. If in a classroom, office or other place where you can lock the door;

- Close and lock your hallway doors. Do not let anyone into the classroom.
- Silence cellphones, darken screens and remain calm & quiet.
- Turn off the lights and shut any available blinds.
- Use desks, tables and other objects to provide protection or barricade doorways.
- Remain sheltered until an All Clear is issued by the Campus Police.

If in an open area;

- Get out of the building immediately using the nearest exit.
- Do not look back or stop until you know you are away and safe.
- Stay out and away from building until and All Clear is issued by the Campus Police.

Tentative Course Calendar: Dates and Assignments may and will be changed as class progress warrants. Actual due dates will be available on Moodle, as are full descriptions of homework assignments.

(Bi)Week	**Topics Covered and Homework Assignments**		
1	Chapter 1 Sections 1.1-1.7	Problems 30 and 31 as flowcharts	MPL: Any 15 questions
	Chapter 2 Sections 2.1-2.17	Programming Challenges 10 and 14	MPL: Any 25 questions
2	Chapter 3 Sections 3.1-3.11	Programming Challenge 12	MPL: Any 25 questions
	Chapter 3 (continued)	Programming Challenges 14 and 25	MPL: Any 25 questions
	EXAM 1 (Chapters 1-3)		
3	Chapter 4 Sections 4.1-4.15	Programming Challenge 12	
	Chapter 4 (continued)	Programming Challenges 14 and 15	MPL: Any 25 questions
4	Chapter 5 Sections 5.1-5.12	Programming Challenge 12	
	Chapter 5 (continued)	Programming Challenges 20 and 25	MPL: Any 25 questions
	EXAM 2 (Chapters 4 and 5)		
5	Chapter 6 Sections 6.1-6.14	Programming Challenge 8	
	Chapter 6 (continued)	Programming Challenges 11 and 12	
6	Chapter 6 (continued)	Programming Challenges 13 and 23	MPL: Any 25 questions
	EXAM 3 (Chapter 6)		
	Chapter 7 Sections 7.1-7.7	Programming Challenge 6-12 with modifications	
7	Chapter 7 (continued)	Programming Challenge 11	MPL: Any 20 questions
	Chapter 8 Sections 8.1-8.4	Programming Challenges 8 and 11	MPL: Any 10 questions
8	Chapter 10 Sections 10.1-10.2, 10.7	Programming Challenges 2 and 12	MPL: Any 10 questions
	EXAM 4 (Chapters 7, 8, and 10)		